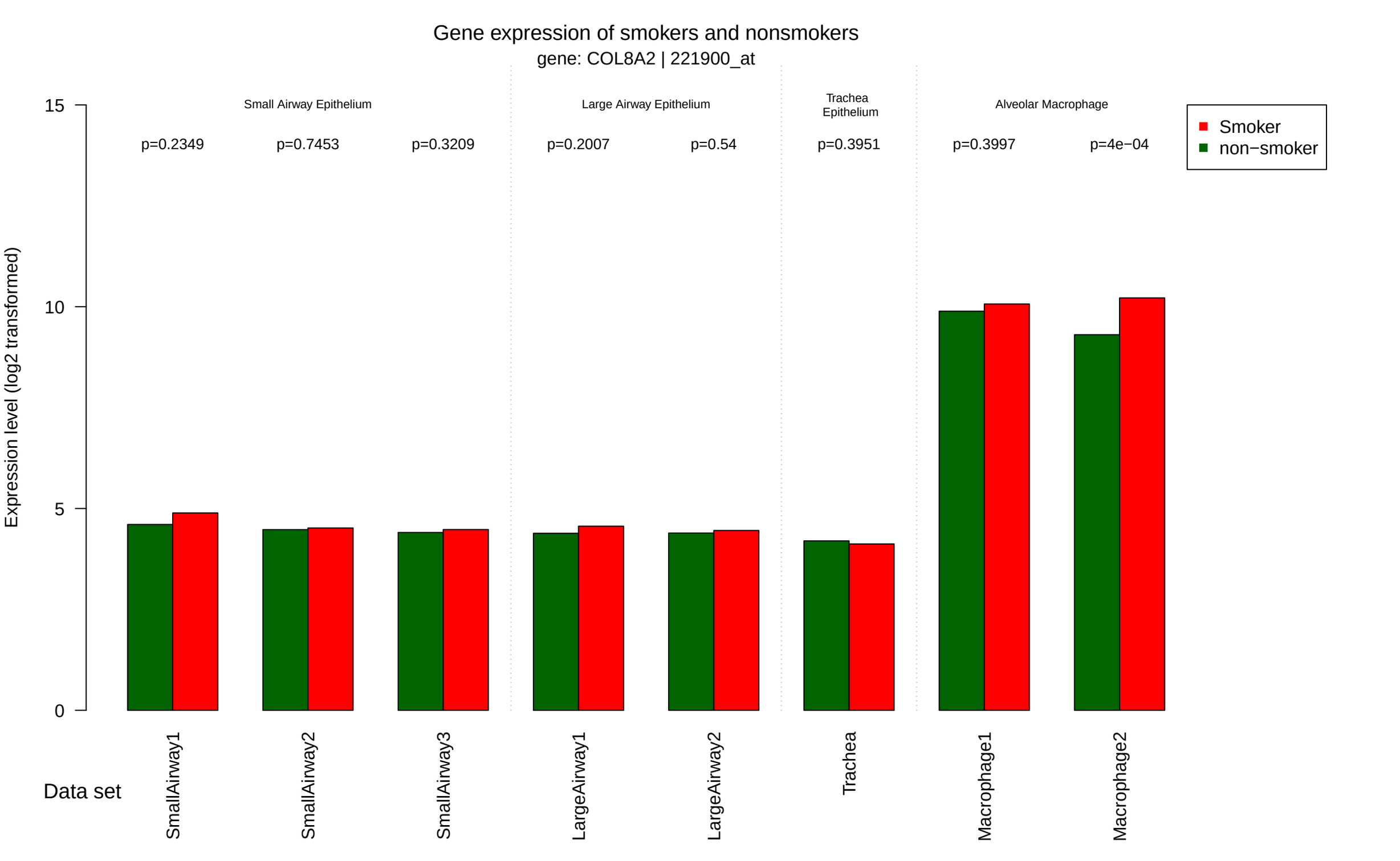
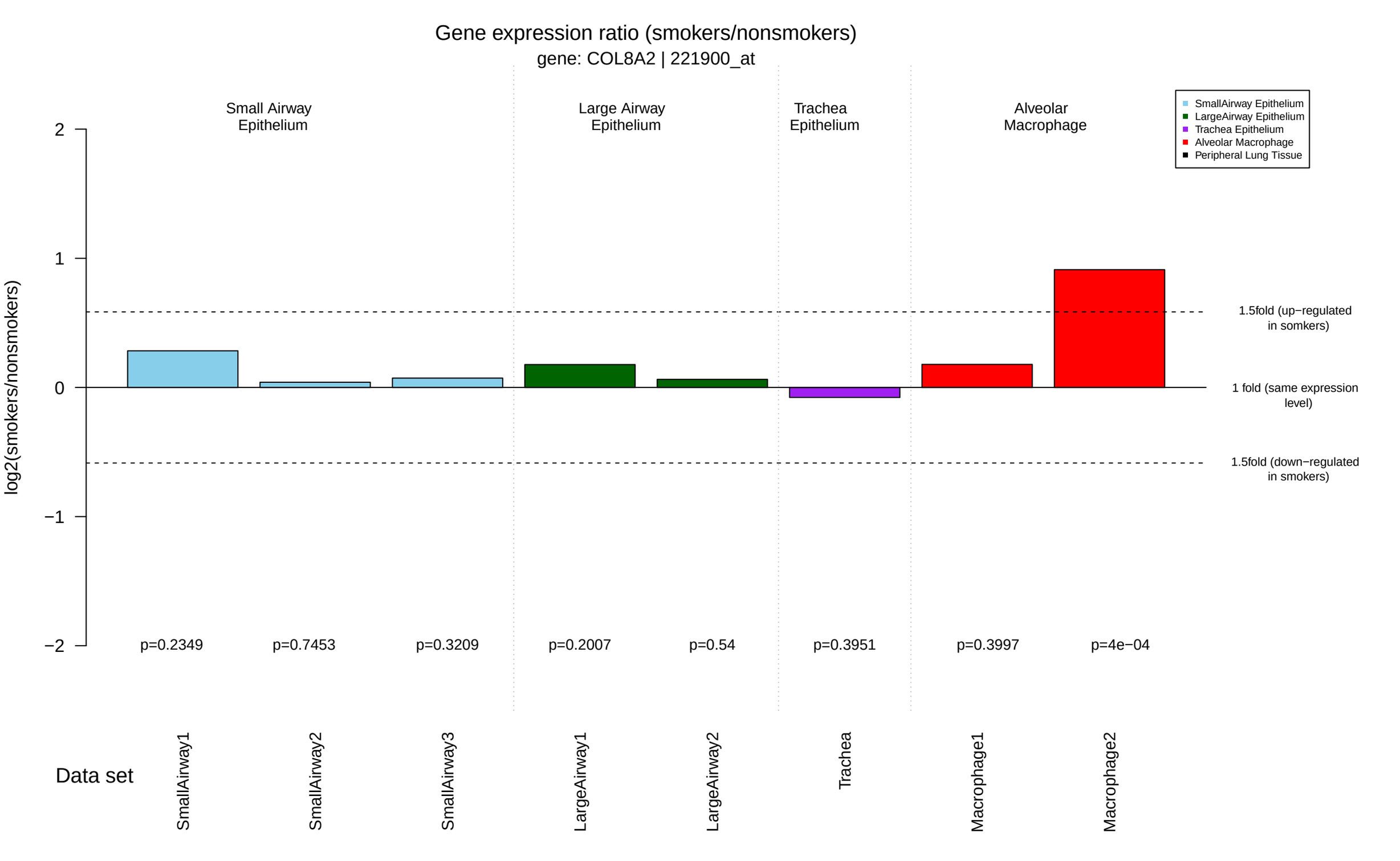
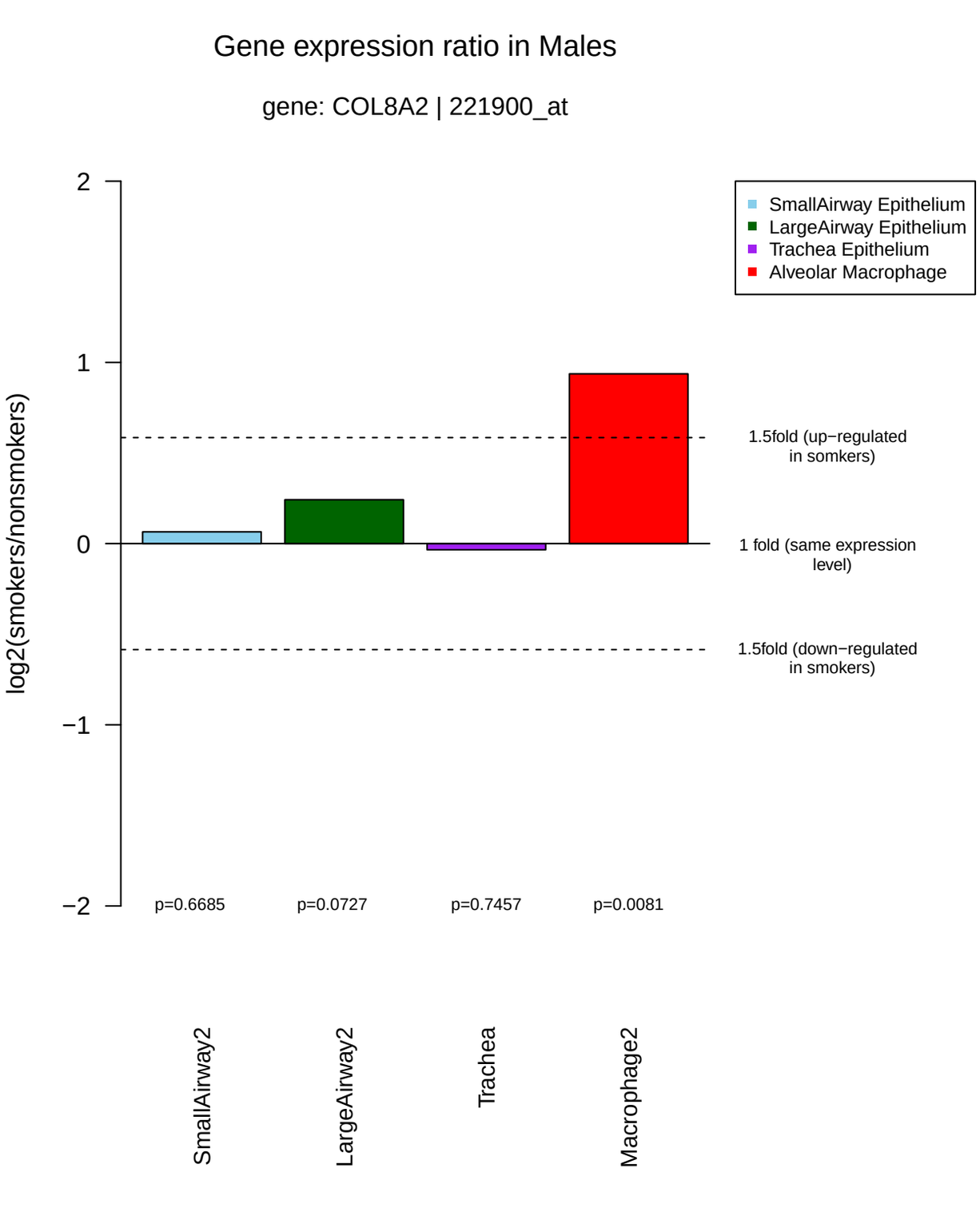
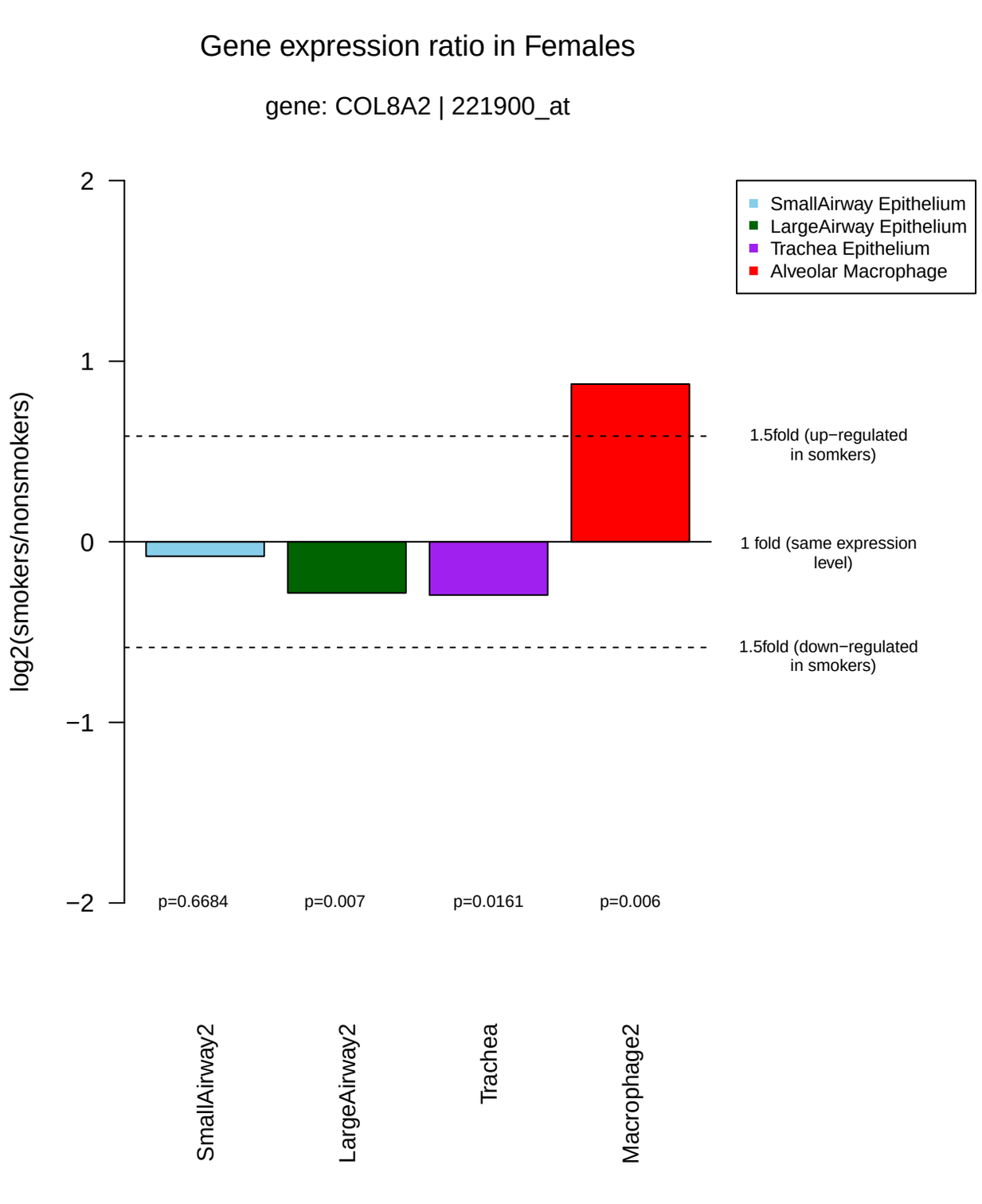
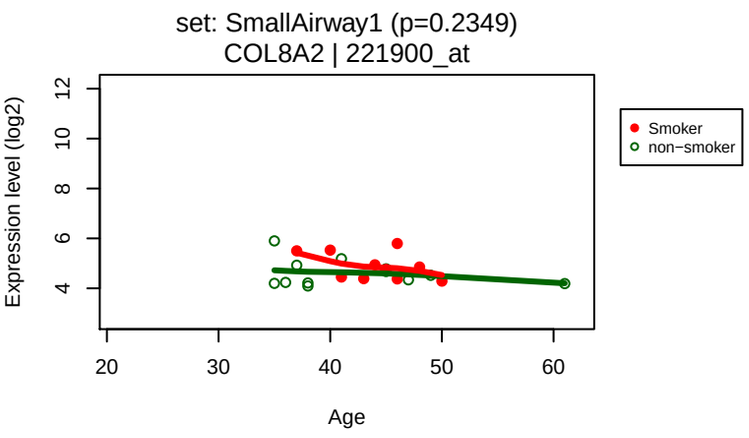
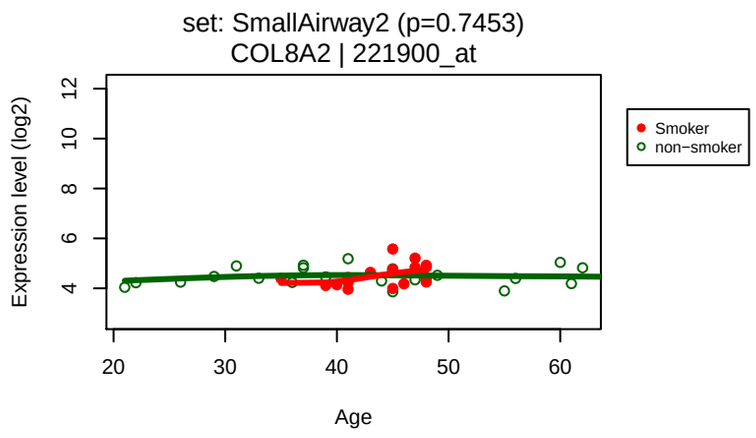

Gene expression of smokers and nonsmokers
gene: COL8A2 | 221900_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2349
p=0.7453
p=0.3209
p=0.2007
p=0.54
p=0.3951
p=0.3997
p=4e−04
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

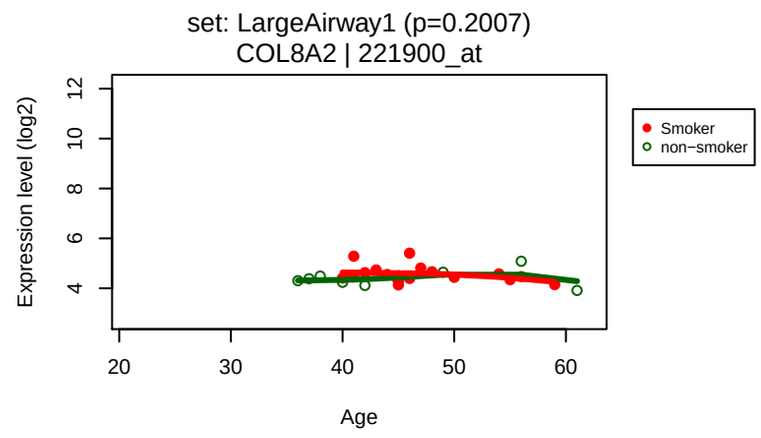

Gene expression ratio (smokers/nonsmokers)
gene: COL8A2 | 221900_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2349
p=0.7453
p=0.3209
p=0.2007
p=0.54
p=0.3951
p=0.3997
p=4e−04
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

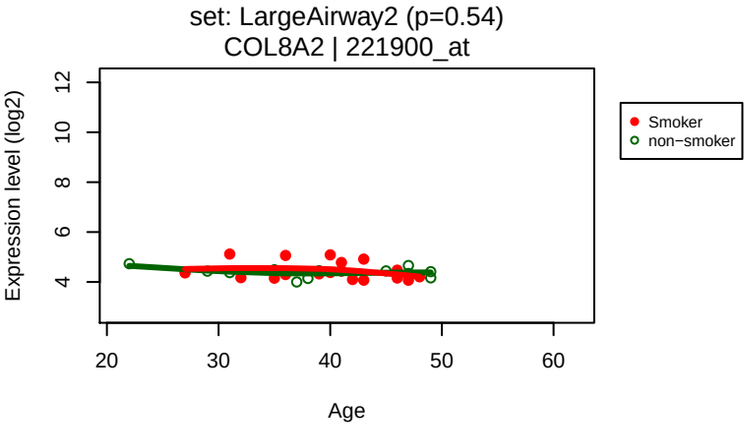

Gene expression ratio in Males
gene: COL8A2 | 221900_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6685
p=0.0727
p=0.7457
p=0.0081
SmallAirway2
LargeAirway2
Trachea
Macrophage2

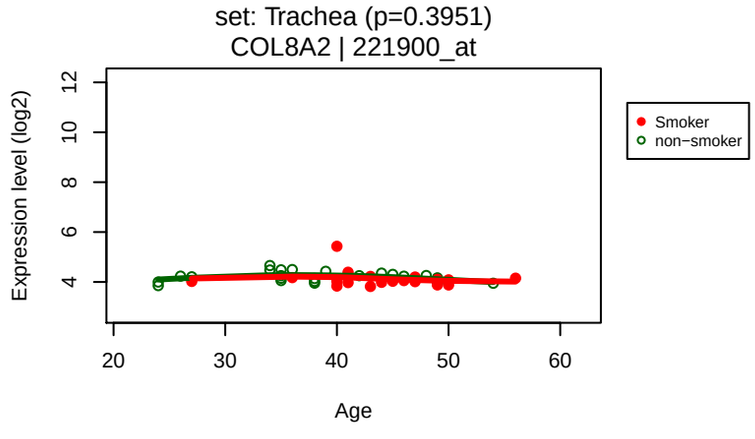

Gene expression ratio in Females
gene: COL8A2 | 221900_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6684
p=0.007
p=0.0161
p=0.006
SmallAirway2
LargeAirway2
Trachea
Macrophage2

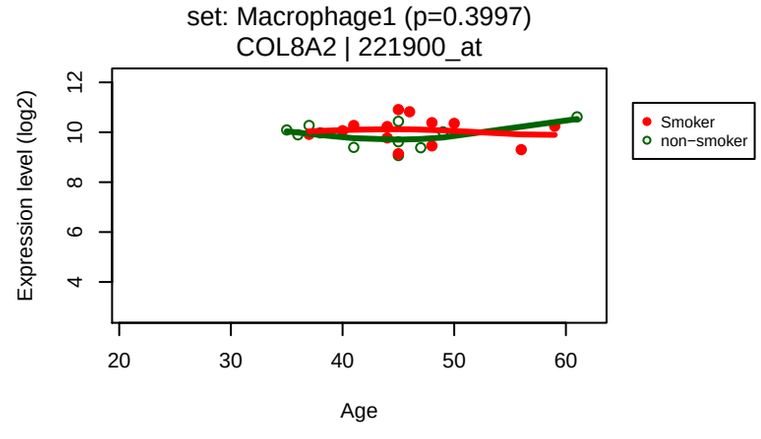

set: SmallAirway1 (p=0.2349)
COL8A2 | 221900_at
Expression level (log2)
Age
Smoker
non−smoker

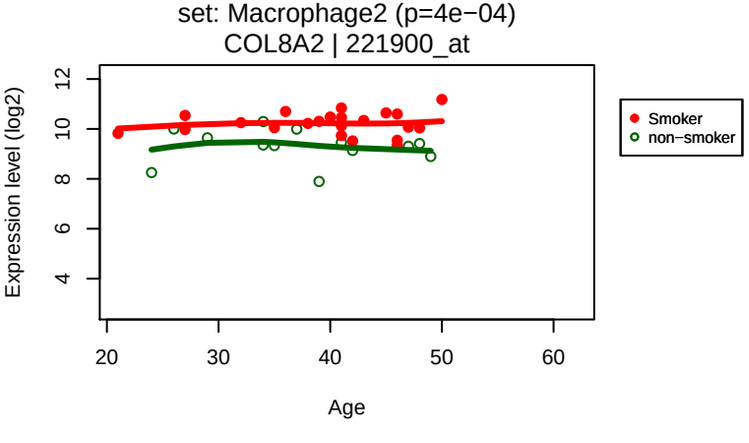

set: SmallAirway2 (p=0.7453)
COL8A2 | 221900_at
Expression level (log2)
Age
Smoker
non−smoker

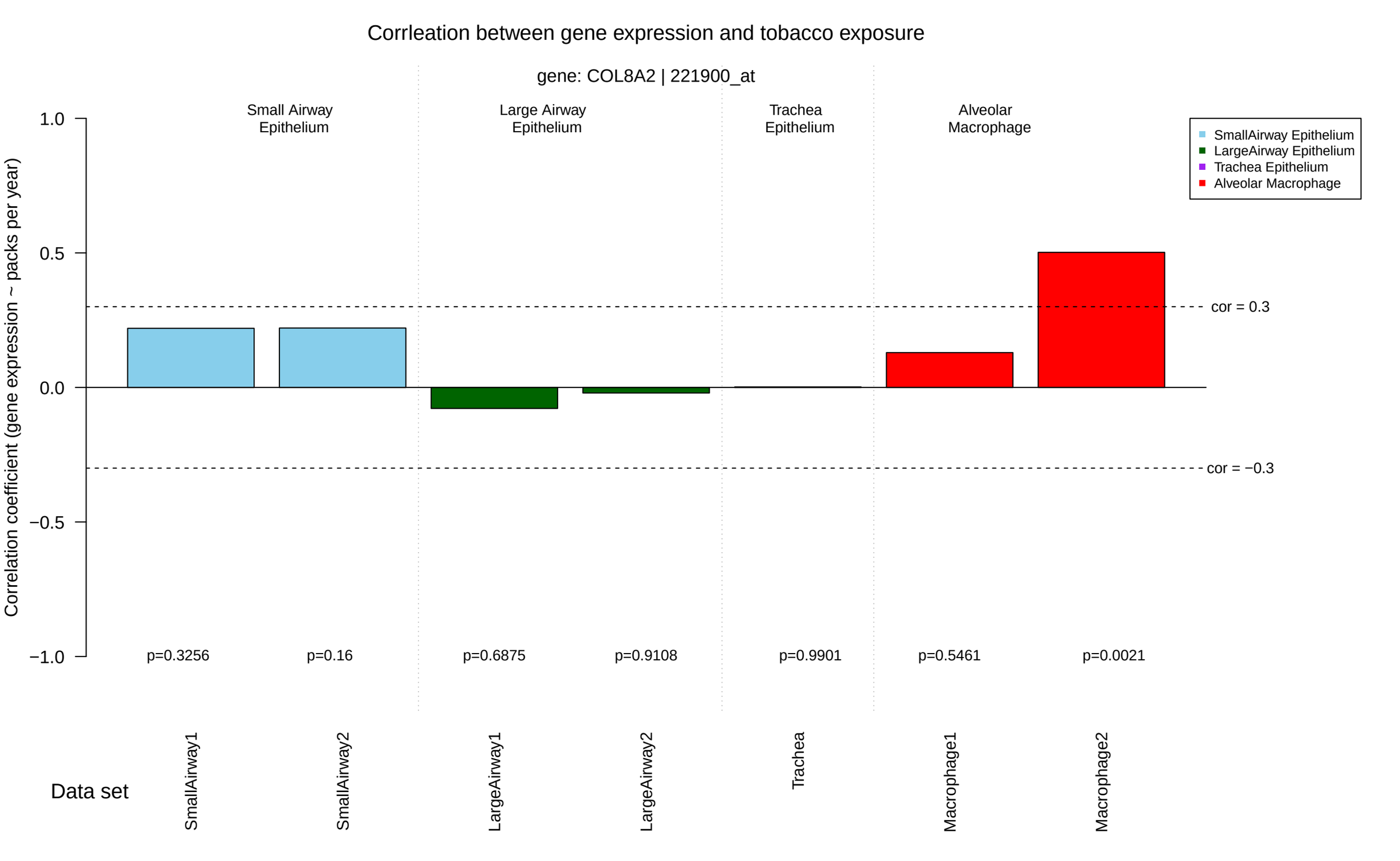

set: LargeAirway1 (p=0.2007)
COL8A2 | 221900_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.54)
COL8A2 | 221900_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3951)
COL8A2 | 221900_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3997)
COL8A2 | 221900_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=4e−04)
COL8A2 | 221900_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL8A2 | 221900_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3256
p=0.16
p=0.6875
p=0.9108
p=0.9901
p=0.5461
p=0.0021
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2